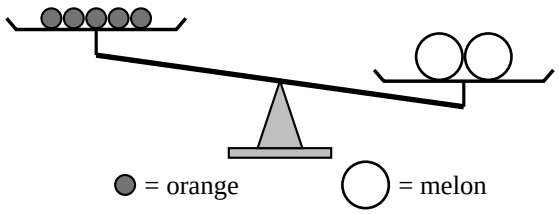
Preview: set-03/by_level/Grade_03-04/y2002_q09 (v1)

Balance scales: remove equal things from both sides, substitute, scale up

Part A: Each question is worth 3 points

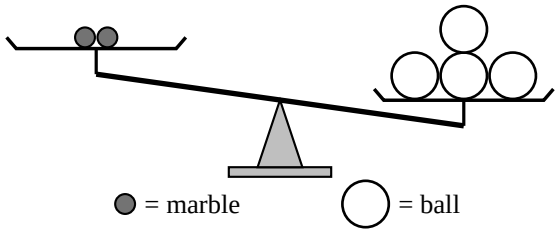
1. On one plate of a balance there are 5 oranges and on the other there are 2 melons. When we put one more melon, exactly like the others, on the plate with the oranges, the balance is level.



One melon weighs the same as how many oranges?

A. 2 oranges ***B.*** 5 oranges ***C.*** 6 oranges ***D.*** 7 oranges ***E.*** 10 oranges

2. On one plate of a balance there are 2 marbles and on the other there are 4 balls. When we put 3 more balls, exactly like the others, on the plate with the marbles, the balance is level.

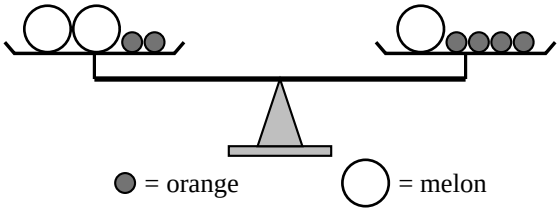


One ball weighs the same as how many marbles?

A. 1 marble ***B.*** 2 marbles ***C.*** 3 marbles ***D.*** 4 marbles ***E.*** 5 marbles

Part B: Each question is worth 4 points

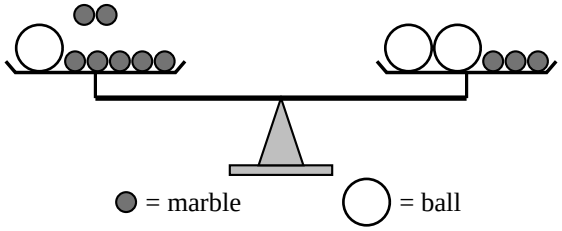
3. The balance is level: 2 melons and 2 oranges on one plate, 1 melon and 4 oranges on the other.



One melon weighs the same as how many oranges?

A. 1 orange ***B.*** 2 oranges ***C.*** 3 oranges ***D.*** 4 oranges ***E.*** 6 oranges

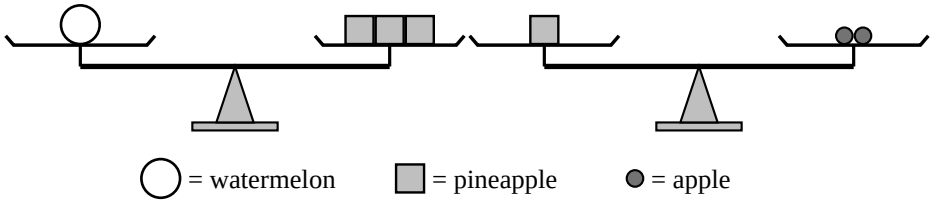
4. The balance is level: 1 ball and 7 marbles on one plate, 2 balls and 3 marbles on the other.



One ball weighs the same as how many marbles?

A. 10 marbles ***B.*** 8 marbles ***C.*** 7 marbles ***D.*** 4 marbles ***E.*** 3 marbles

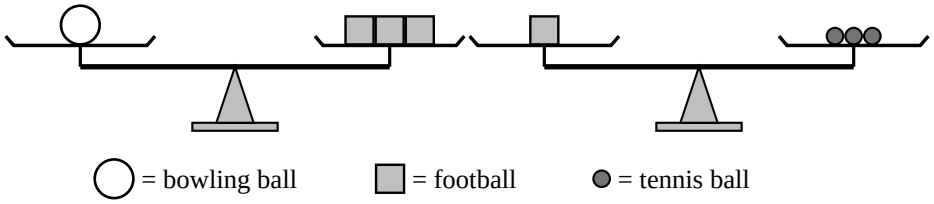
5. The two balances are level. One watermelon weighs as much as 3 pineapples, and one pineapple weighs as much as 2 apples.



How many apples weigh as much as 1 watermelon?

A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 6

6. The two balances are level. One bowling ball weighs as much as 3 footballs, and one football weighs as much as 3 tennis balls.

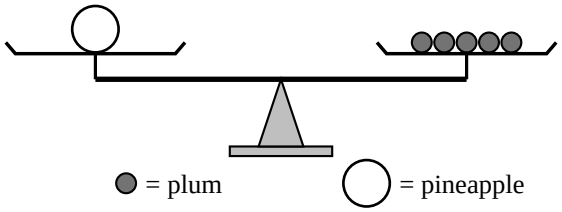


How many tennis balls weigh as much as 2 bowling balls?

A. 6 ***B.*** 9 ***C.*** 10 ***D.*** 12 ***E.*** 18

Part C: Each question is worth 5 points

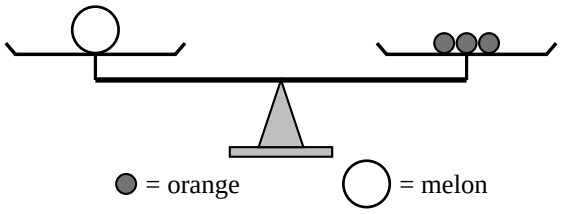
7. The balance shows that one pineapple weighs as much as 5 plums. 2 pineapples and 4 plums weigh 700 g altogether.



How much does one plum weigh?

A. 25 g ***B.*** 50 g ***C.*** 100 g ***D.*** 250 g ***E.*** 450 g

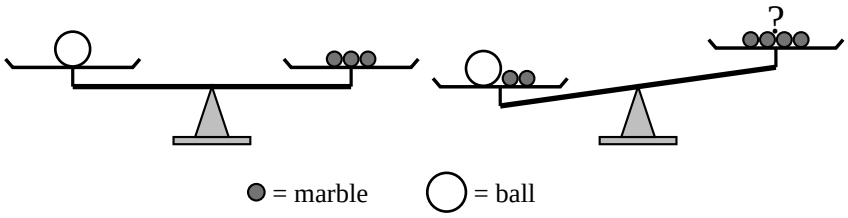
8. The balance shows that one melon weighs as much as 3 oranges. 3 melons and 2 oranges weigh 1650 g altogether.



How much does one orange weigh?

A. 150 g ***B.*** 125 g ***C.*** 100 g ***D.*** 75 g ***E.*** 50 g

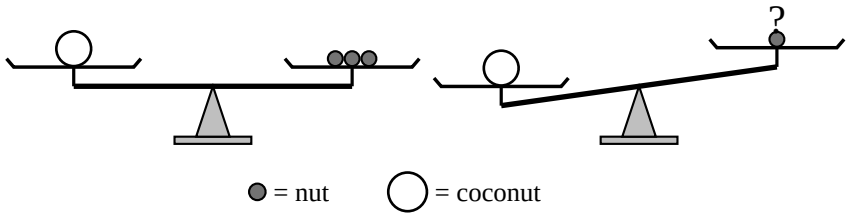
9. The first balance is level. On the second balance there is 1 ball and 2 marbles on the left and 4 marbles on the right.



How many marbles must be added to the right plate to make it level?

A. 1 ***B.*** 2 ***C.*** 4 ***D.*** 5 ***E.*** 9

10. The first balance is level. On the second balance there is 1 coconut on the left and 1 nut on the right.



How many nuts must be added to the right plate to make it level?

A. 6 ***B.*** 5 ***C.*** 4 ***D.*** 3 ***E.*** 2

Answer Key

1.	**B**	**2.**	**B**	**3.**	**B**	**4.**	**D**	**5.**	**E**	**6.**	**E**
7.	**B**	**8.**	**A**	**9.**	**A**	**10.**	**E**				

Solutions

1. B (5 oranges) — Take 1 melon off both plates: 5 oranges balance 2 – 1 = 1 melon. So one melon = 5 oranges. [set-03/by_level/Grade_03-04/2002 Q9 · one_more_balances]

2. B (2 marbles) — Take 3 balls off both plates: 2 marbles balance 4 – 3 = 1 ball. So one ball = 2 marbles. [set-03/by_level/Grade_03-04/2002 Q9 · one_more_balances]

3. B (2 oranges) — Take 1 melon and 2 oranges off both plates. Left: 1 melon = 2 oranges. [set-03/by_level/Grade_03-04/2002 Q9 · both_sides]

4. D (4 marbles) — Take 1 ball and 3 marbles off both plates. Left: 1 ball = 4 marbles. [set-03/by_level/Grade_03-04/2002 Q9 · both_sides]

5. E (6) — One watermelon = 3 pineapples = 3 × 2 = 6 apples. [set-03/by_level/Grade_03-04/2002 Q9 · chain]

6. E (18) — One bowling ball = 3 footballs = 3 × 3 = 9 tennis balls; 2 bowling balls = 2 × 9 = 18 tennis balls. [set-03/by_level/Grade_03-04/2002 Q9 · chain]

7. B (50 g) — Swap each pineapple for 5 plums: 2 × 5 + 4 = 14 plums weigh 700 g. One plum: 700 ÷ 14 = 50 g. [set-03/by_level/Grade_03-04/2002 Q9 · grams]

8. A (150 g) — Swap each melon for 3 oranges: 3 × 3 + 2 = 11 oranges weigh 1650 g. One orange: 1650 ÷ 11 = 150 g. [set-03/by_level/Grade_03-04/2002 Q9 · grams]

9. A (1) — One ball = 3 marbles, so the left plate is worth 1 × 3 + 2 = 5 marbles. The right plate has 4, so add 5 – 4 = 1. [set-03/by_level/Grade_03-04/2002 Q9 · what_to_add]

10. E (2) — One coconut = 3 nuts, so the left plate is worth 1 × 3 = 3 nuts. The right plate has 1, so add 3 – 1 = 2. [set-03/by_level/Grade_03-04/2002 Q9 · what_to_add]